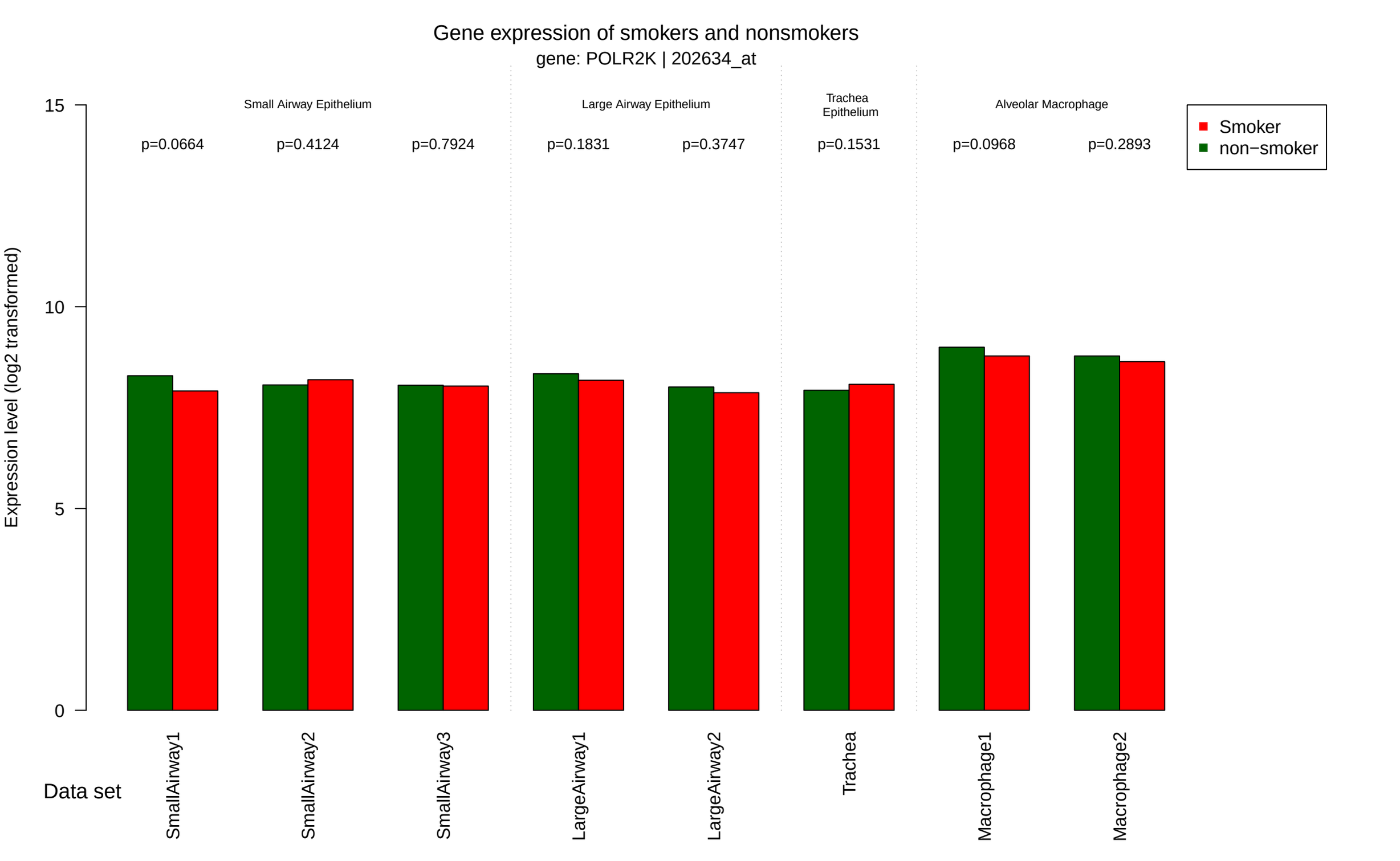
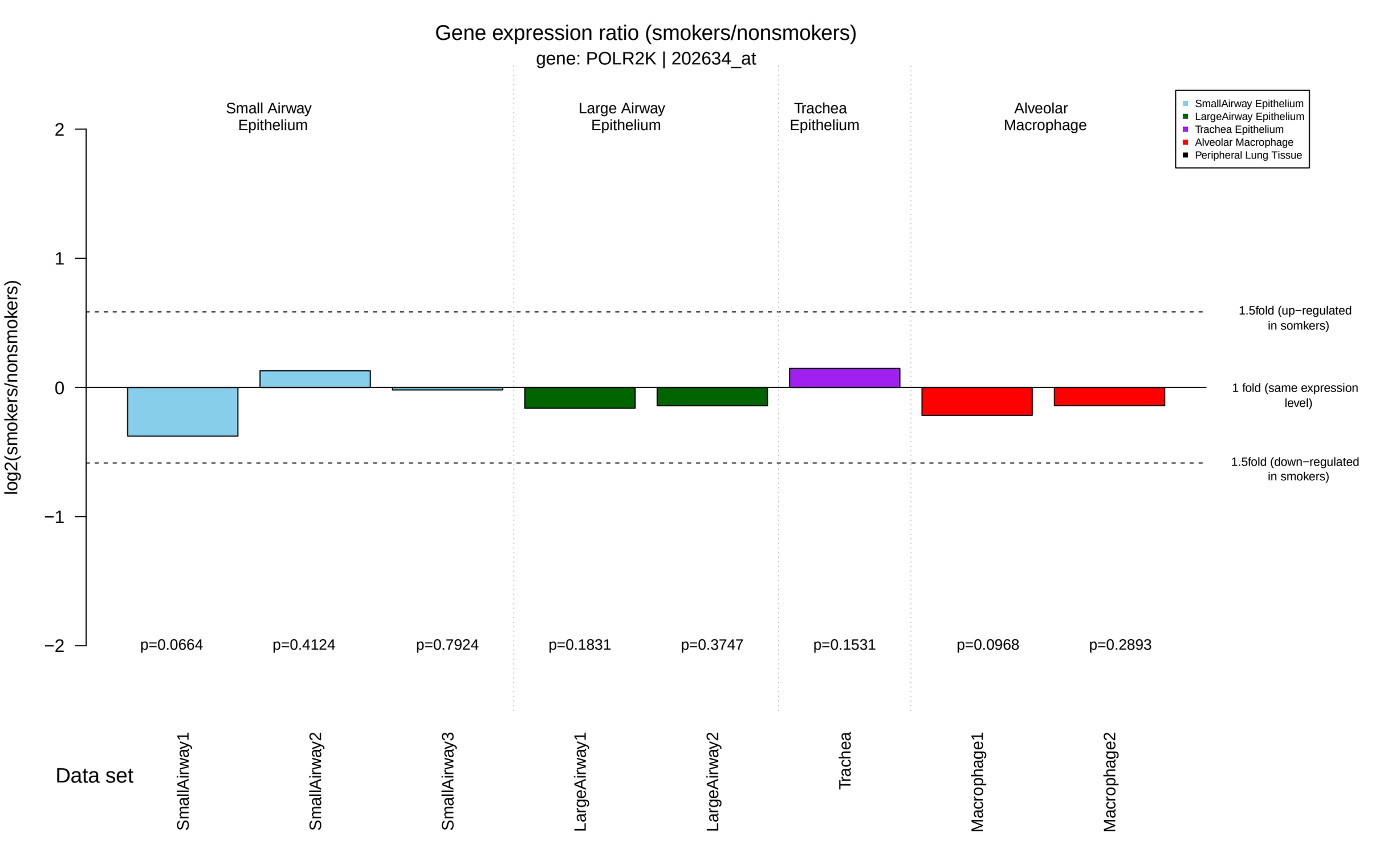
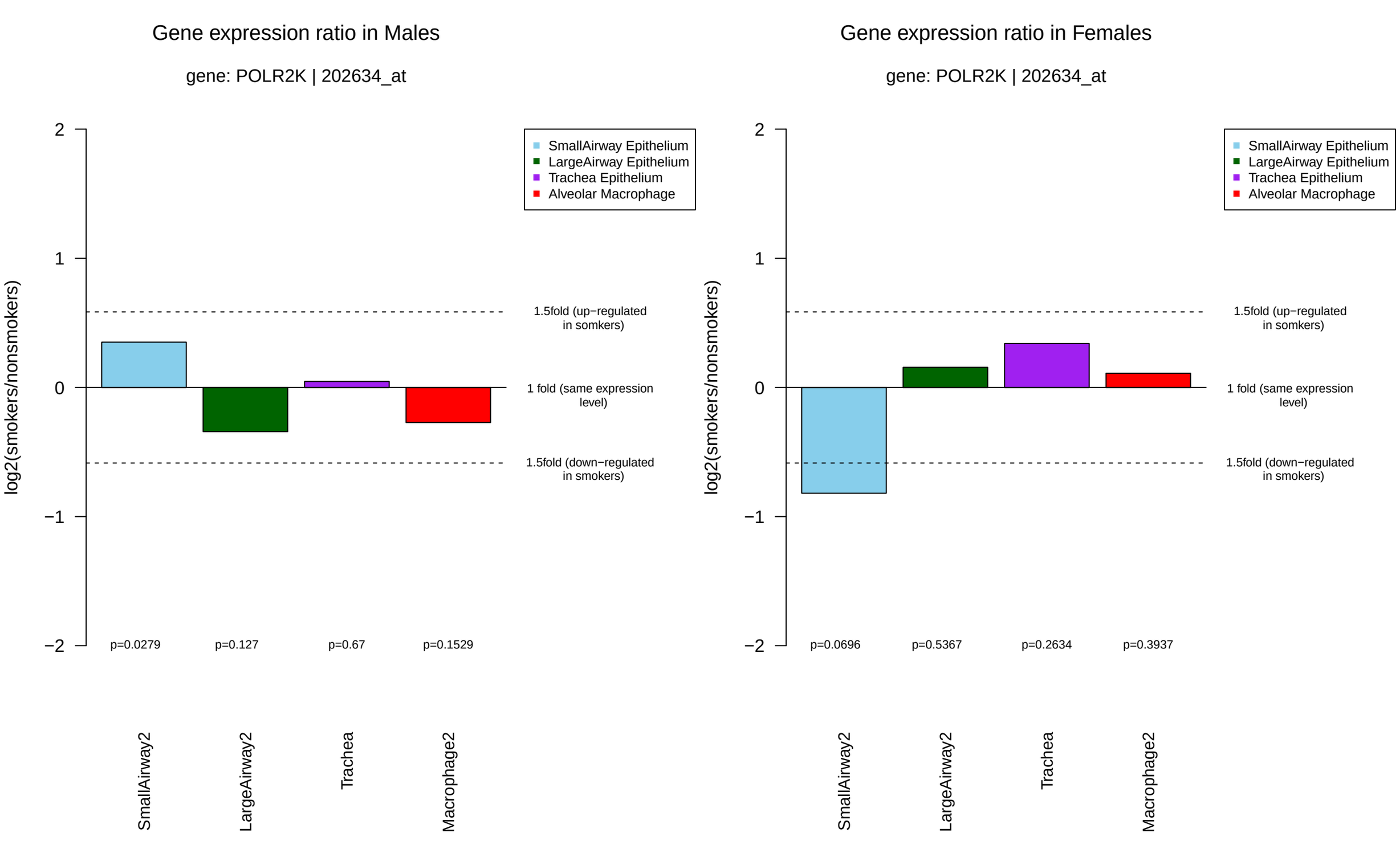
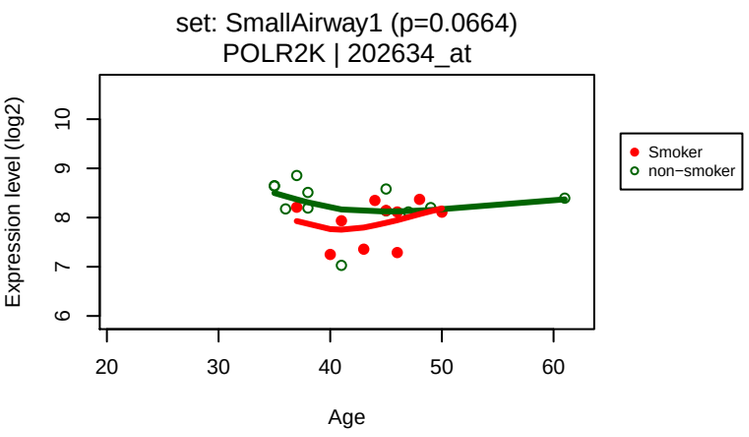
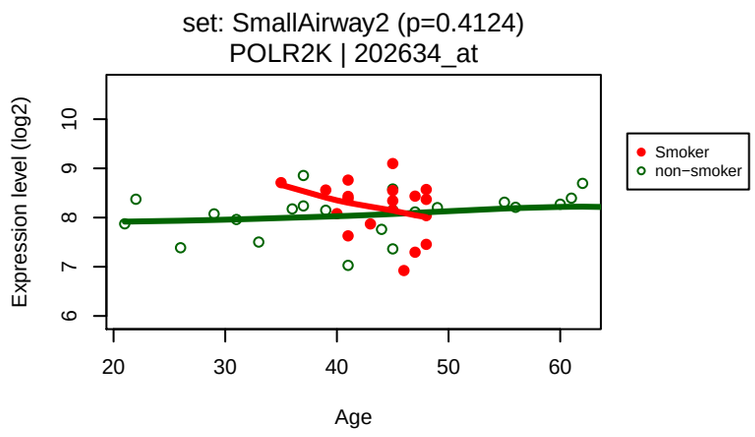
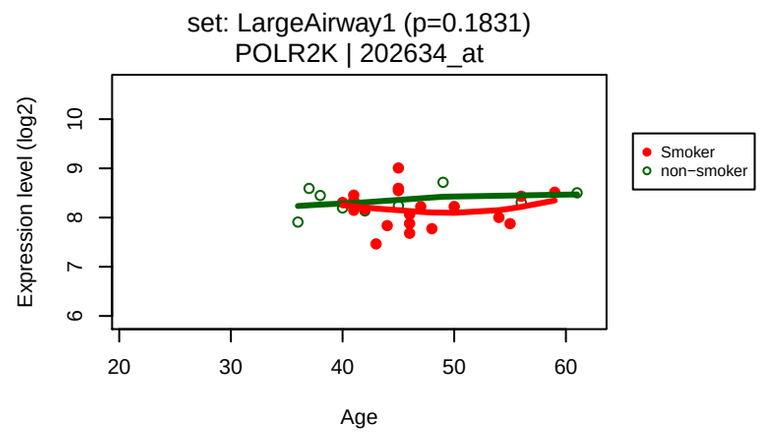

Gene expression of smokers and nonsmokers
gene: POLR2K | 202634_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0664
p=0.4124
p=0.7924
p=0.1831
p=0.3747
p=0.1531
p=0.0968
p=0.2893
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

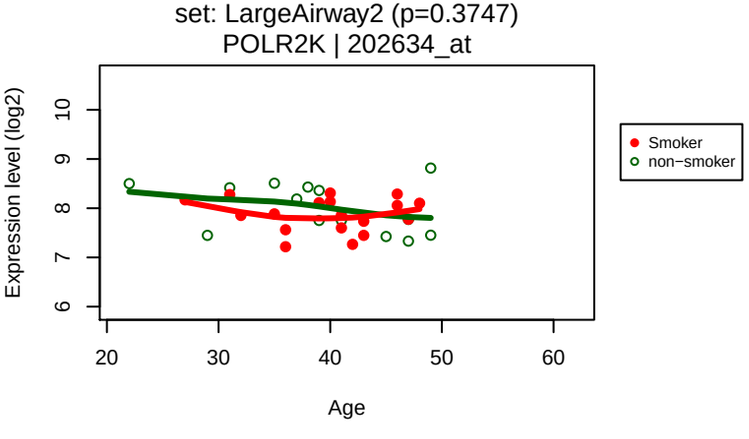

Gene expression ratio (smokers/nonsmokers)
gene: POLR2K | 202634_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0664
p=0.4124
p=0.7924
p=0.1831
p=0.3747
p=0.1531
p=0.0968
p=0.2893
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

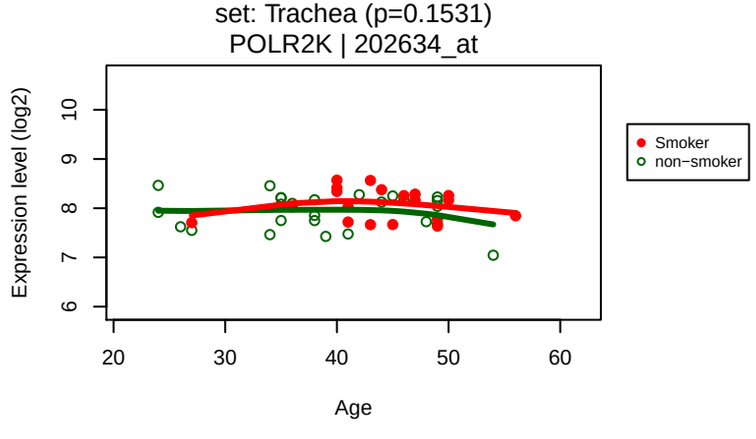

Gene expression ratio in Males
gene: POLR2K | 202634_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0279
p=0.127
p=0.67
p=0.1529
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR2K | 202634_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0696
p=0.5367
p=0.2634
p=0.3937
SmallAirway2
LargeAirway2
Trachea
Macrophage2

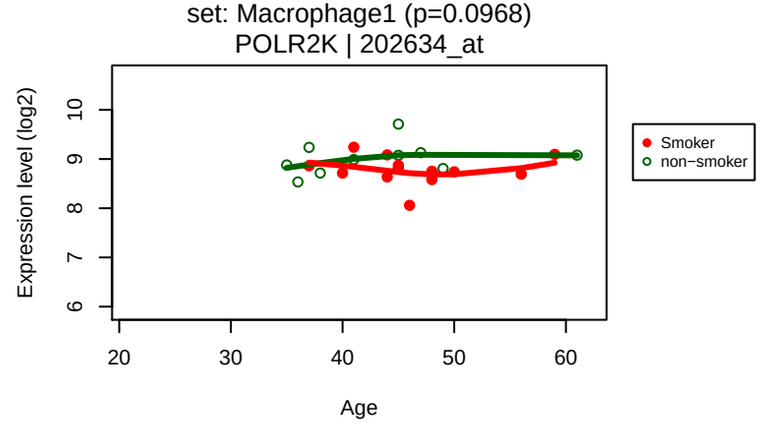

set: SmallAirway1 (p=0.0664)
POLR2K | 202634_at
Expression level (log2)
Age
Smoker
non−smoker

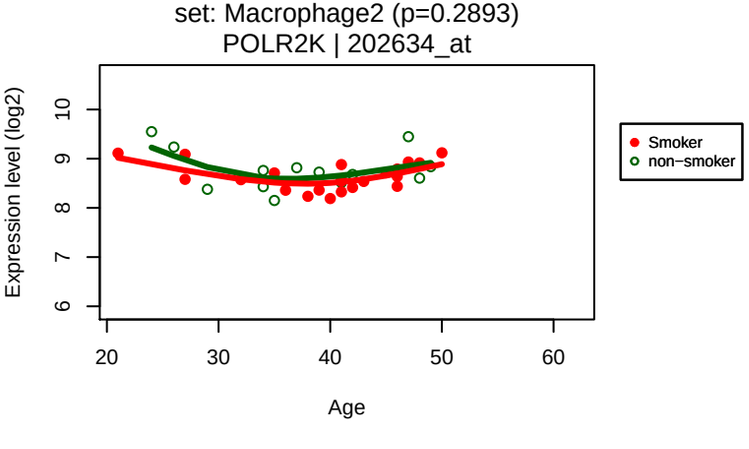

set: SmallAirway2 (p=0.4124)
POLR2K | 202634_at
Expression level (log2)
Age
Smoker
non−smoker

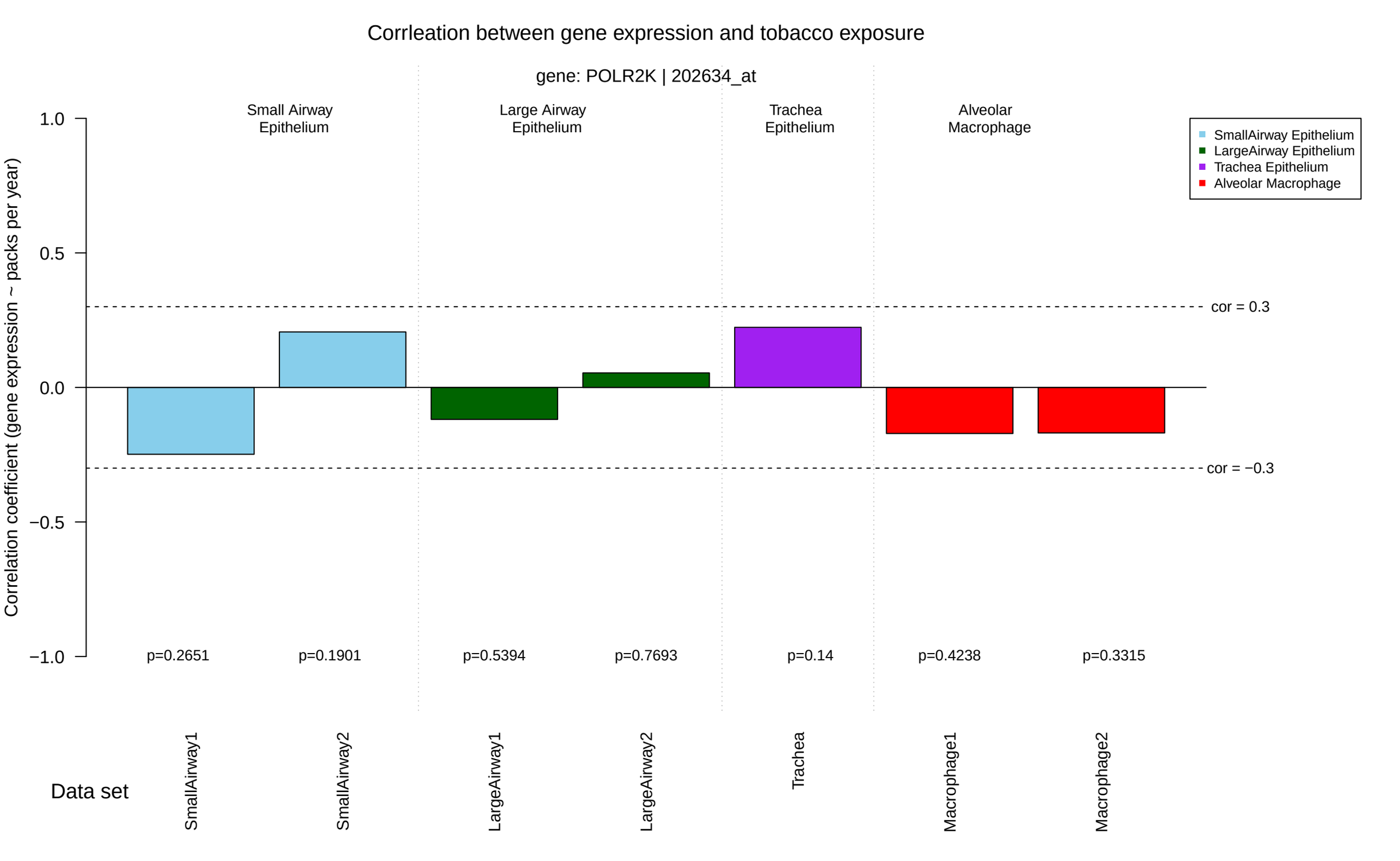

set: LargeAirway1 (p=0.1831)
POLR2K | 202634_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3747)
POLR2K | 202634_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1531)
POLR2K | 202634_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0968)
POLR2K | 202634_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2893)
POLR2K | 202634_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR2K | 202634_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2651
p=0.1901
p=0.5394
p=0.7693
p=0.14
p=0.4238
p=0.3315
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2